

Feeding behavior recognition of group-housed pigs based on pose estimation and keypoint features discrimination

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ABSTRACT

In the field of intelligent sensing for smart animal husbandry, accurate recognition of feeding behavior in group-housed pigs is crucial for achieving precision farming and improving pig welfare. Currently, pig feeding behavior recognition relies on detection boxes-based methods, which are difficult to exclude Non-Nutritive Visiting Behavior within the feeding zone. To precisely recognize the feeding behavior of group-housed pigs, this study proposes a feeding behavior recognition method based on pose estimation and keypoint features discrimination. Firstly, Pig-HRNet is designed to estimate the pose of group-housed pigs, in which a Context Transformer (COT) attention module is specially introduced to detect the keypoints of pigs more accurately under crowded conditions. Secondly, by analyzing the correlation between keypoints and feeding zone, group-housed pigs are divided into visiting the feeding zone and Non-Feeding Behavior (NFB). For visiting the feeding zone, the behaviors are further categorized into Feeding Behavior (FB) and Non-Nutritive Visiting Behavior (NNVB). The experimental data of group-housed pigs were collected in commercial pig farms, including a total of 1400 video frames. Experimental results show that the Pig-HRNet model achieves an average precision (AP) of 97.1% in estimating pig poses. Compared to other pose estimation network models such as KAPAO, HigherHRNet, DeepLabCut, and HRNet, the detection AP improved by 69.0%, 16.3%, 12.3%, and 0.5%, respectively. The feeding behavior recognition method proposed in this paper achieved precision and recall rates of 98.8% and 99.9%, respectively. The relevant results indicate that the proposed feeding behavior recognition method performs well, while also meeting the requirement for accurately estimating pig poses under crowded conditions. The feeding behavior dataset established in this paper has been shared on <https://github.com/IPCLab-NEAU/Group-housed-pigs-Feeding-Behavior-Recognition> for use by the precision animal husbandry research community.

1. Introduction

In recent years, with the expansion of the pig farming industry, the farming practices have shifted from traditional methods to precision farming (Tzanidakis et al., 2021). The premise of precision farming is monitoring the behaviors of group-housed pigs (Monteiro et al., 2021). Observing the feeding behavior of pigs enables understanding their eating patterns and intake quantities, allowing timely manual intervention to be adopted when abnormal feeding behaviors occur in pigs (Mao et al., 2023; Matthews et al., 2016). Currently, in the process of pig farming, monitoring of pig feeding behavior relies mainly on manual

observation, which has low automation levels and cannot provide continuous and precise watching of pigs (Arizono et al., 2010; Fang et al., 2021; Loreille and Bouchet, 2003). Therefore, it is significant for precision farming and maintaining pig welfare to employ intelligent methods to recognize feeding behavior of pigs.

In recent years, researchers have started using intelligent perception technology to recognize feeding behaviors of pigs, primarily employing methods based on sensors and computer vision technology. In terms of sensor-based methods, (Garrido-Izard et al., 2020) determine the identity of the pig visiting the feeding zone through its unique RFID tag, and monitored feeding behavior by measuring changes in food before and

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after the pig's visit. (Kapun et al., 2020) tagged pigs with UHF-RFID ear tags on their ears. When the UHF-RFID reader above the feeding zone detects the signal from the UHF-RFID ear tag, it identifies the pig's behavior as feeding. The accuracy of feeding behavior identification is approximately 80 %. However, sensor-based methods not only increase breeding costs but also cause harm to pigs. Particularly in group-housing environments, large-scale deployment of sensors markedly raises breeding expenses, and pigs frequently attempting to remove sensors often result in device damage, further increasing maintenance and replacement costs. Especially when using ear-tag sensors, the tagging process may cause ear tissue injury, and pigs' attempts to tear off the tags can also lead to ear damage.

With the development of computer vision technology, it has become possible to use this technology for non-contact and low-stress feeding behavior recognition in group-housed pigs. Common computer vision technology-based methods for pig feeding behavior recognition can be divided into two categories: methods based on individual pig detection boxes and pig head detection boxes.

In the individual pig detection box-based method, researchers annotate the different behaviors of group-housed pigs and detect its behaviors using object detection models. (Li et al., 2023) integrated the Squeeze-and-Excitation (SE) attention module into the YOLOX model, and establish the detection model SE-YOLOX, which is used to detect the feeding behavior of pigs. The model achieved a feeding behavior recognition mean average precision of 88.03 %. (Tran and Thanh, 2023) employed the YOLOv7 model to detect and recognize pigs' stationary behaviors: standing, lying, and feeding. The precision and recall rates for pig feeding behavior reached 97 %. (Tu et al., 2024) used YOLOv5 to detect individual pig behaviors (lying, standing, feeding, other), achieving a feeding behavior recognition precision of 91.8 % and a recall rate of 89.9 %. From the monitoring perspective, the head of the pig holds greater significance for recognizing its feeding behavior. However, the method based on individual pig detection boxes does not focus on pig head information, often resulting in misidentification of feeding behavior.

To more accurately recognize pig feeding behavior, researchers determine pig feeding behavior by checking if the pig's head is in the feeding zone or not (Maselyne et al., 2015). In the pig head detection box-based method, (Yang et al., 2018) employed the FRPD (Faster R-CNN pig detector) model to detect pig heads. Feeding behavior occurs when the bounding box of the head intersects with the feeding zone. The precision and recall rates for feeding behavior recognition are 99.6 % and 86.93 %, respectively. (Chen et al., 2021) utilized a combination of Xception and LSTM models to recognize feeding behavior of pigs. Feeding behavior is identified when the ratio of the pig head detection boxes to the feeding zone is greater than 10 % in each frame, and the accumulated pixels of head motion is greater than 60 in the 1 s episode. (Kim et al., 2021) employed three object detection algorithms, YOLOv4, YOLOv3, and improved YOLOv3, to detect the heads of pigs in feeding zone. Through comparison, they found that using the YOLOv4 algorithm yielded the best detection results. Feeding behavior is recognized when the bounding box of a pig's head intersects with the feeding zone, achieving an average precision of 89.16 % for feeding behavior recognition. (Zhuang et al., 2023) cropped the captured images to retain only the head areas of pigs in the feeding zone, designating these pigs as feeding pigs. Annotate the cropped images and the feeders without pigs present. Using the MobileNetV2 model, they classified the labeled images, achieving a feeding behavior recognition recall rate higher than 97 %. (Luo et al., 2024) proposed the tph-YOLOv5 model for detecting pig heads, recognize feeding behaviors based on the ratio of the bounding box of pig heads to the feeding zone. Although the pig head detection box-based method takes into account the head information of pigs, it cannot identify cases where the head is in the feeding zone but the pig is not actually feeding.

In real farming environments, the specific behavior of a pig's mouth being in the feeding zone but not feeding is referred to as Non-Nutritive

Visiting Behavior (NNVB). The aforementioned pig feeding behavior recognition methods mainly rely on object detection boxes, making it difficult to accurately distinguish between Feeding Behavior (FB) and NNVB. (Alameer et al., 2020) transformed the pig feeding recognition task into a classification task. They cropped the images to retain only the pig head regions within the feeding zone and used the Sc-GoogLeNet classification network to classify the cropped images into seven categories: 1 FB, 1 FB 1 NNVB, 1 NNVB, 2 FB, 2 FB 1 NNVB, 2 NNVB, and None (where none of the above scenarios occurred). However, this method cannot identify the individual behavior of each pig in the image.

To accurately recognize the feeding behavior of individual pigs in group-housed, this study proposes a feeding behavior recognition method based on pose estimation and keypoint features discrimination. Firstly, a pose estimation network, Pig-HRNet, is proposed to detect keypoints of individual pigs in a group-housed environment. Secondly, the behavior of group-housed pigs in images is classified into visiting the feeding zone and Non-Feeding Behavior (NFB) based on the relationship between the Nose keypoint and the feeding zone. Thirdly, clustering is performed on pigs visiting the feeding zone using features such as the angle formed by the Left_ear-Nose-Right_ear and the Euclidean distance between the Nose and Shoulder. The optimal hyperplane between different categories is calculated and used as the criterion for determining FB and NNVB. The main contributions of this paper are as follows:

1. A pose estimation network, Pig-HRNet, is proposed to address the issue of misdetection of individual keypoints in group-housed pigs under crowded conditions.
2. By combining pose estimation and keypoint features discrimination, Feeding Behavior (FB), Non-Nutritive Visiting Behavior (NNVB), and Non-Feeding Behavior (NFB) were accurately recognized.
3. A dataset for group-housed pig feeding behavior was created. The dataset is publicly available on <https://github.com/IPCLab-NEAU/Group-housed-pigs-Feeding-Behavior-Recognition>.

The paper is organized into four sections: Section 1 introduces the research background and significance, current research status, existing problems, and the main contributions of this paper. Section 2 describes the data construction, the methods used in this study, and the experimental environment. Section 3 presents the experimental results and provides an in-depth analysis and discussion. Section 4 summarizes the entire paper and provides an outlook for future work.

2. Materials and methods

2.1. Data acquisition

The experimental video samples were collected in October 2018 at Harbin Hongfu Pig Farm. The pig pen was 4.3 m long and 2.3 m wide. Each pig pen contained 2 feeding zones and housed 10 hybrid piglets. The age of the pigs is 35–42 days old, there is no sex differentiation. Considering that the recently weaned piglets still had the habit of free feeding, a Hikvision DS-2CD3345D-I model camera was used to record continuous video footage of these 10 hybrid piglets for 72 h to ensure an adequate number of feeding behaviors in the videos. The camera was positioned 2.3 m above the ground. The recorded group-housed pig videos were stored in MP4 format with a resolution of 2560 × 1440 pixels and a frame rate of 25 fps.

2.2. Dataset

In the recorded videos of group-housed pigs, pigs typically maintain relatively static postures (such as keeping their heads down to feed) for several minutes during feeding episodes. To avoid data duplication and redundancy, frames were extracted from the recorded videos every 300 s. This sampling interval is sufficient to capture Feeding Behavior (FB),

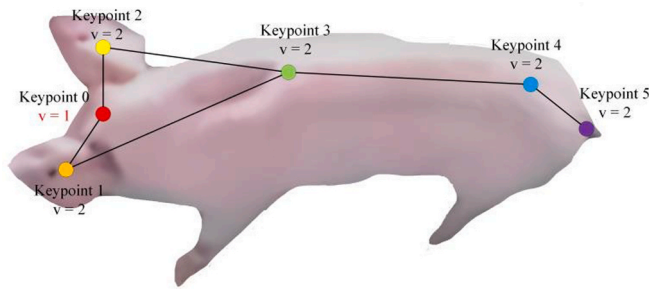


Fig. 1. Keypoints and keypoints visibility(v) annotation. Keypoint 0: Nose; Keypoint 1: Left ear; Keypoint 2: Right ear; Keypoints 3: Shoulder; keypoints 4: Buttocks; Keypoint 5: Tail.

Table 1

Custom keypoints and connection lines.

Keypoints ID	Custom keypoints	Connection lines
0	Nose	(Nose, Left ear) (Nose, Right ear)
1	Left ear	(Left ear, Shoulder)
2	Right ear	(Right ear, Shoulder)
3	Shoulder	(Shoulder, Buttocks)
4	Buttocks	(Buttocks, Tail)
5	Tail	

Non-Nutritive Visiting Behavior (NNVB), and Non-Feeding Behavior (NFB). During the extraction process, images with motion blur or unclear targets caused by pig movement were manually deleted. The remaining images were randomly sorted, resulting in a total of 1400 RGB images of group-housed pigs.

The 1400 images were annotated using the COCO-Annotator annotation tool to establish a dataset for group-housed pig pose estimation. This scheme includes 6 custom keypoints for individual group-housed pigs, as well as keypoint connection lines. Keypoint 0 represents the snout of the pig, which serves as the starting point for the pig skeleton diagram. Keypoint 1 and 2 represent the center points of the left and right ears of the pig, respectively. Keypoint 3 is a surface point on the central spine of the back, between the shoulder blades. Keypoint 4 is another surface point on the central spine of the back, located between the left and right hindquarters. Keypoint 5 represents the connection between the pig's tail and hindquarters, specifically the base of the tail. Despite using an overhead perspective to reduce occlusion, the natural clustering behavior of group-housed pigs may still lead to individual occlusion in images. Therefore, during annotation, keypoint visibility (v) is set to 2 for clearly visible points and 1 for occluded but present points. Different loss weights are applied during training accordingly. Fig. 1 shows annotation examples.

The connection lines link the custom keypoints of the group-housed pigs, indicating the relative positional relationship between the connected keypoints. The specific relationship between each keypoint and the connection lines is shown in Table 1.

The annotated 1400 images will be randomly divided into training and testing sets in a 7:3 ratio to create a group-housed pig pose estimation dataset.

Based on data from the animal ethology literature (Alameer et al., 2020; Luo et al., 2024) and under the guidance of animal science experts, the individual behaviors of pigs in 1400 images were categorized

into three types: Feeding Behavior (FB), Non-Nutritive Visiting Behavior (NNVB), and Non-Feeding Behavior (NFB). The definitions of the three behaviors are shown in Table 2.

These behaviors were organized to construct a group-housed pig feeding behavior dataset. For subsequent experiments, the constructed pig feeding behavior dataset was also split in a 7:3 ratio and aligned with the pose estimation dataset. Statistics on the different behaviors of pigs in the training and test sets were compiled, as shown in Table 3.

2.3. Overall architecture

The overall architecture of the group-housed pig feeding behavior recognition method is shown in Fig. 2, which outlines the process of recognize feeding behavior in group-housed pigs, consisting of two stages: pig pose estimation and keypoint features discrimination.

In the pig pose estimation stage, the Pig-HRNet network detects keypoints of group-housed pigs. Six keypoints of individual pigs were detected, labeled as Nose, Left ear, Right ear, Shoulder, Buttocks, Tail. In the keypoint features discrimination stage, Firstly, pig behaviors are classified into visiting the feeding zone and Non-Feeding Behavior (NFB) based on the relationship between the Nose keypoint and the feeding zone. Secondly, the angle formed by the Left ear-Nose-Right ear and the Euclidean distance between the Nose and Shoulder are used as features to cluster the pigs visiting the feeding zone. The optimal hyperplane between different categories learned from clustering is then used as the criterion for determining Feeding Behavior (FB) and Non-Nutritive Visiting Behavior (NNVB) of the pigs.

2.3.1. Group-housed pig pose estimation based on Pig-HRNet

The first step of the proposed group-housed pig feeding behavior recognition method is to perform pose estimation on the pigs to obtain keypoint coordinate information. Due to the unique living habits of pigs, they often appear crowded in images. To accurately locate pig keypoints under crowded conditions, a network called Pig-HRNet is proposed, which integrates the CotNet Block into the high-resolution network HRNet. Compared to common pose estimation models (Cheng et al., 2020; Mathis et al., 2018; McNally et al., 2022), Pig-HRNet maintains high-resolution features of the image through parallel multi-resolution paths and enhances the network's representational ability by fusing information from different resolutions. The CotNet Block (Li et al., 2022) enhances the expressiveness of the output features by focusing on the characteristics of local regions and global contextual information,

Table 2

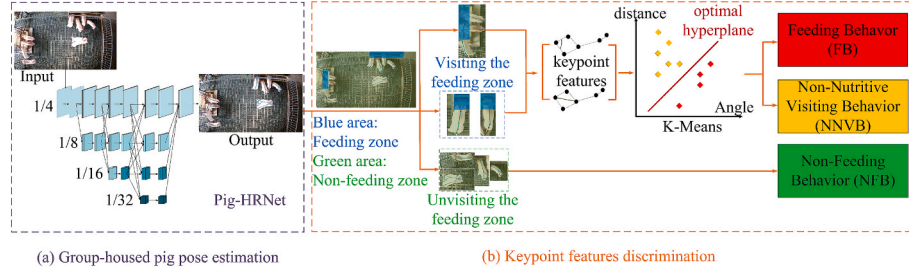
Definition and description of feeding behavior of pigs.

Behavior category	Abbreviation	Description
Feeding Behavior	FB	The pig's mouth is inside the feeding zone, its head is lowered, and both forelimbs enter or are close to the feeding zone.
Non-Nutritive Visiting Behavior	NNVB	The pig's mouth is within the feeding zone, but it never consumed any food.
Non-Feeding Behavior	NFB	The pig does not approach the feeding zone, its mouth is not inside the feeding zone, its body is close to or far from the feeding zone.

Table 3

Counting the number of pigs for three behaviors.

	Training Set	Testing Set
Number of FB Pigs	1339	801
Number of NNVB Pigs	505	115
Number of NFB Pigs	7948	2655

**Fig. 2.** The overall process of the group-housed pig feeding behavior recognition method.

allowing it to infer the positions of occluded keypoints from the surrounding visible parts.

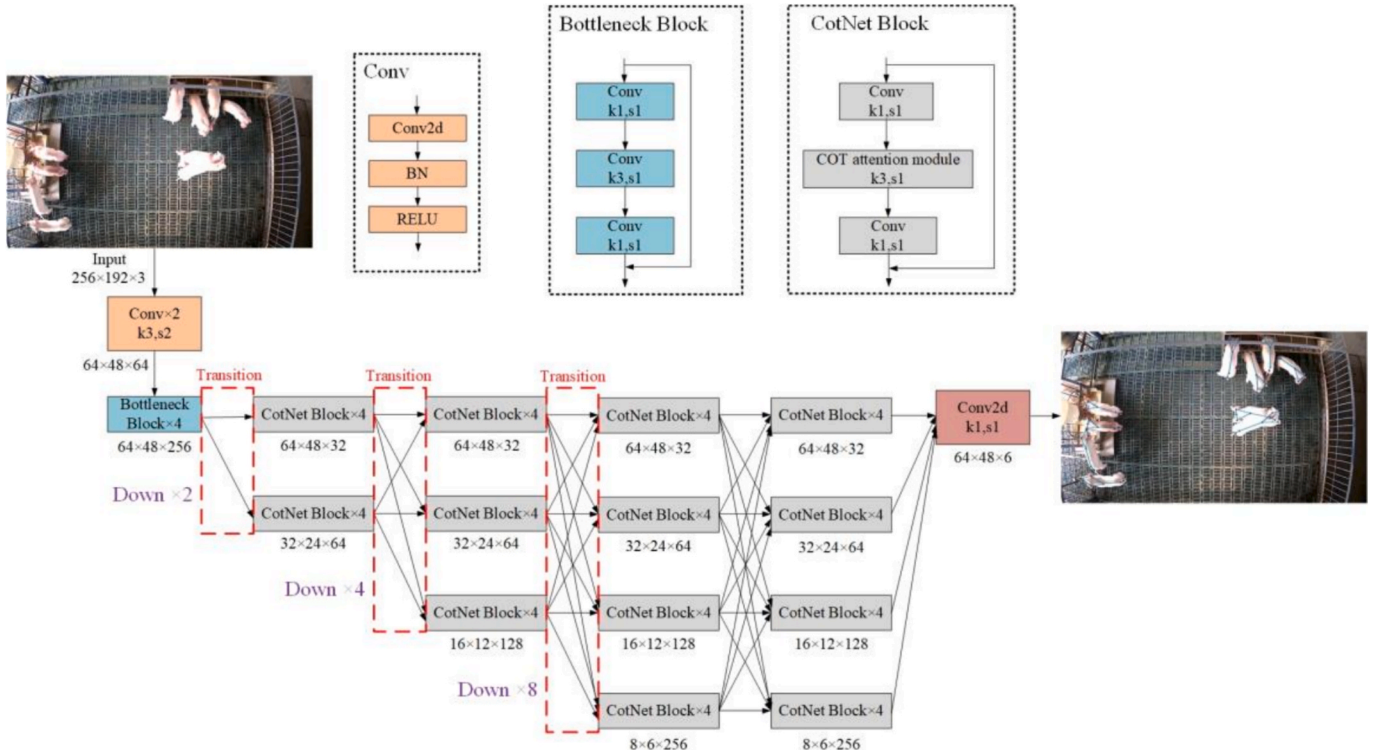
The structure of Pig-HRNet is illustrated in Fig. 3. Pig-HRNet utilizes parallel branches at multiple resolutions and facilitates information interaction between different branches to achieve strong semantic information and precise spatial information simultaneously. This approach overcomes the limitation of traditional sequential networks by preserving both semantic and spatial information through the use of parallel branches and inter-branch information exchange.

Firstly, the input image is processed using two convolutional layers with a kernel size of 3×3 and a stride of 2 to convert the RGB image into feature maps. The feature maps are subjected to preliminary information extraction using the Bottleneck Block module, which is repeated four times. The size of the feature maps remains unchanged, while the channel size is adjusted to 256.

48 through upsampling by different factors, and feature fusion is performed. This layer has dimensions of $64 \times 48 \times 6$, corresponding to the heatmap for each keypoint.

This study introduces the CotNet Block to enhance the network's learning of both global and local information in the images, addressing the issue of incorrect keypoint detection in group-housed pigs under crowded conditions.

The CoNet Block consists of two 1×1 convolutions and a Context Transformer (COT) attention module. The COT attention module effectively inherits the global contextual information from traditional self-attention while also incorporating the local contextual information brought by convolution, thus enhancing visual representation capabilities. The backbone network of HRNet is composed of the Basic Block structure, which has a small receptive field during feature learning. Traditional self-attention can capture global information to obtain a

**Fig. 3.** Pig-HRNet network architecture.

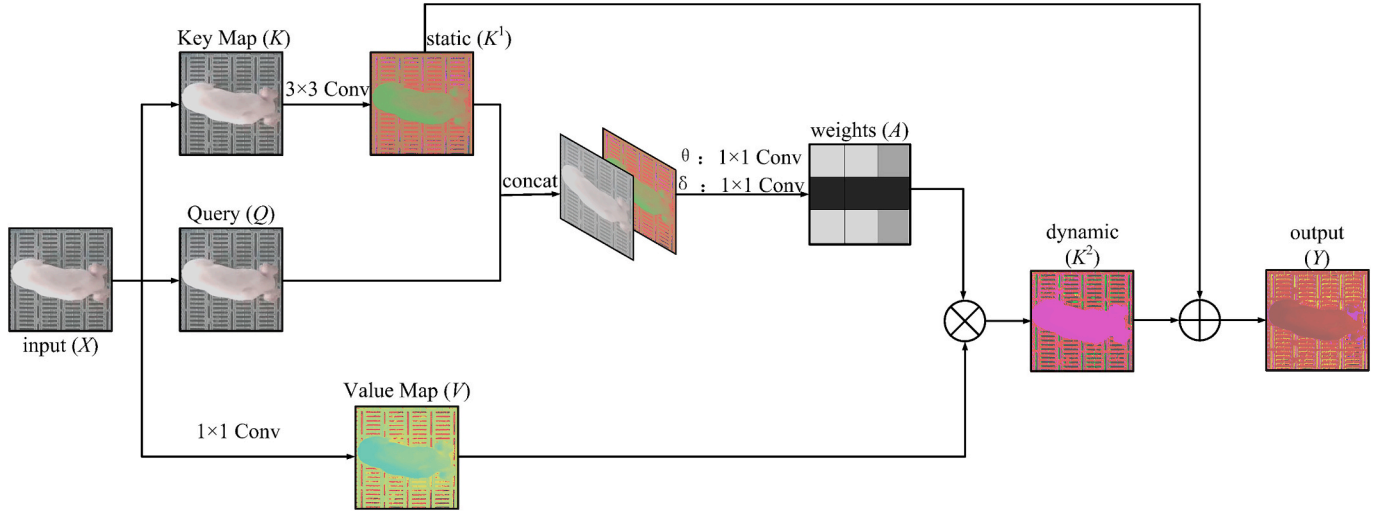


Fig. 4. Context transformer (COT) attention module.

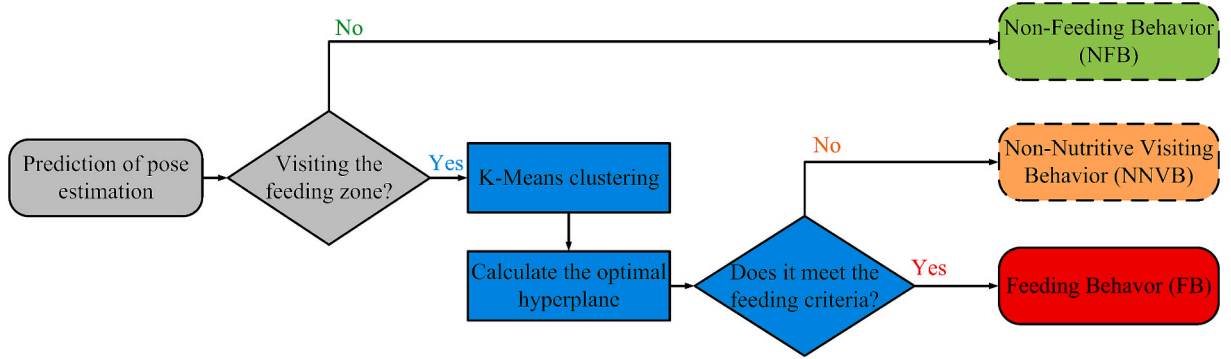


Fig. 5. Feeding behavior determination of group-housed pigs.

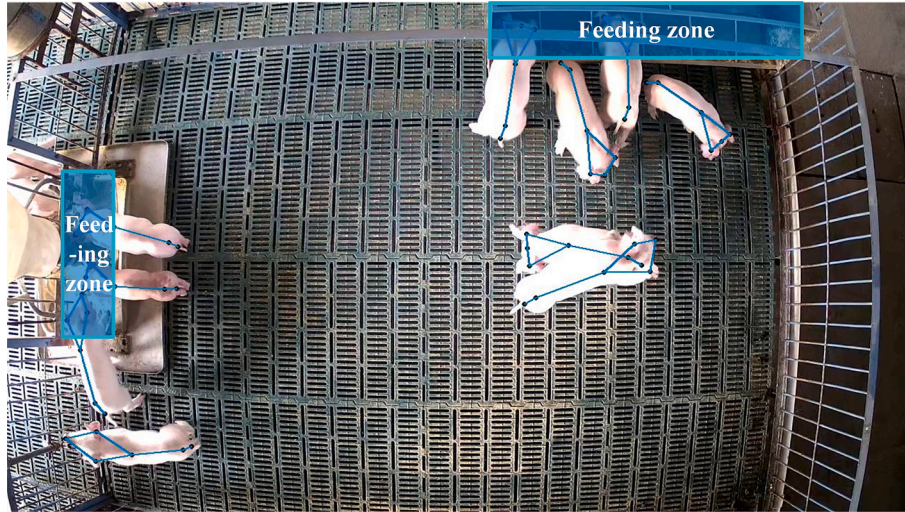


Fig. 6. Feeding zone in the monitoring piggy.

larger receptive field but overlooks rich local contextual information. The COT attention module integrates static contextual information with dynamically obtained contextual information through attention mechanisms into a unified framework. The structure of the COT attention module is shown in Fig. 4.

For the input feature X , three variables are defined: Key Map (K), Query (Q), and Value Map (V). Only the feature mapping for V is carried

out, while Q and K retain their original X values, i.e., $K = X$, $Q = X$, $V = XW_v$. A group convolution of size 3×3 is applied to K to obtain K^1 , which represents local contextual information. K^1 is then concatenated with Q , and is passed through two consecutive 1×1 convolutions W_θ and W_δ to generate the attention matrix, termed Attention Map (A):

$$A = [K^1, Q] W_\theta W_\delta \quad (1)$$

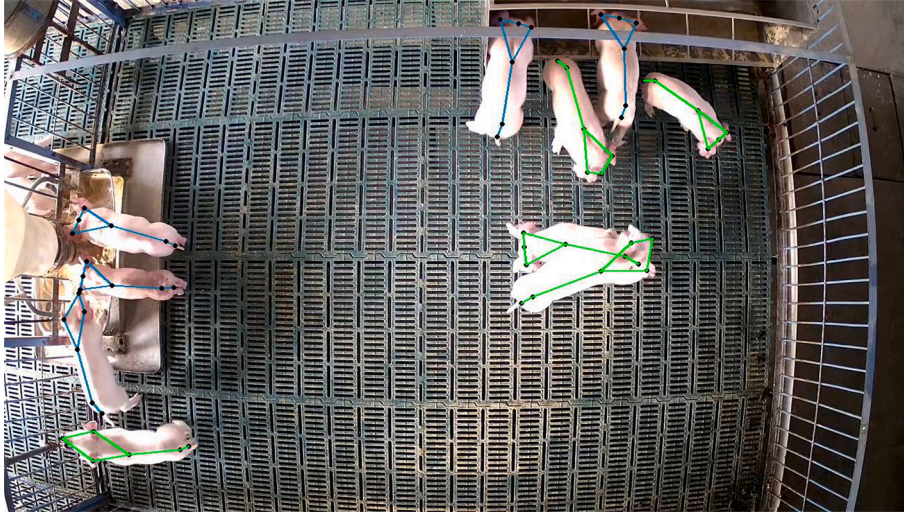


Fig. 7. Visiting the feeding zone.

The matrix A is obtained by the interaction between Q and the local contextual information K^1 , enhancing the self-attention mechanism. A is then multiplied element-wise with V to calculate the attention feature map, termed K^2 :

$$K^2 = V \otimes A \quad (2)$$

The output Y is the fusion of K^1 , which models local static context, and K^2 , which models global dynamic context.

2.3.2. Feeding behavior recognition

Currently, the recognition of feeding behavior in group-housed pigs is generally based on the object detection boxes, which determines whether feeding has occurred by assessing the overlap between the head box of the group-housed pigs and the feeding zone. In this study, Pig-HRNet is used to estimate the pose of group-housed pigs, obtaining keypoint coordinate information, and employing clustering and optimal hyperplane to recognize the feeding behavior of pigs. The process of feeding behavior determination is illustrated in Fig. 5.

The feeding behavior of pigs is correlated with certain functional area. The feeding behavior of group-housed pigs is manifested in the image as Lowing head (Lou and Hurnik, 1998). Keeping head in the feeding zone, biting and chewing on its food, and rooting with snout in the feeder (Lao et al., 2016; Maselyne et al., 2015). In order to monitor the feeding behavior of pigs, we define feeding zone in our monitoring scene. As shown in Fig. 6, the 2 blue rectangle represents the feeding zone.

(1) Recognition of visiting the feeding zone and NFB.

During feeding, the mouth area of pigs must be within the feeding zone. This article accurately determines the Nose keypoint of group-housed pigs through Pig-HRNet, further narrowing down the range of contact between pigs and food. By judging whether the coordinates of the Nose keypoint are within the feeding zone, pigs are classified into visiting the feeding zone pigs and NFB pigs. As shown in Fig. 7, pigs visiting the feeding zone are represented by blue skeletons, while NFB pigs are represented by green skeletons.

(2) Recognition of FB and NNVB.

A pig is considered to be feeding when its mouth is inside the feeding zone and its head is lowered. A Non-Nutritive Visiting Behavior (NNVB) was scored when a pig entered the feeding zone without ever consuming any food. As shown in Fig. 8, the yellow rectangle represents the NNVB, while the blue rectangle represents the Feeding Behavior (FB). During feeding in the feeding zone, pigs often exhibit a head-down posture, resulting in a relatively large angle formed by the Left_ear-Nose-Right_ear, and a relatively small Euclidean distance between the Nose and Shoulder. Pigs exhibiting NNVB often have a head-up posture, leading to a relatively small angle formed by the Left_ear-Nose-Right_ear, and a relatively large Euclidean distance between the Nose and Shoulder.

For pigs visiting the feeding zone, classification should first be performed using the K-Means clustering method (Coates and Ng, 2012). The angle α formed by the Left_ear-Nose-Right_ear and the Euclidean distance l between the Nose and Shoulder of each pig are used as keypoint features. The keypoint features are shown in Fig. 8, The red curve

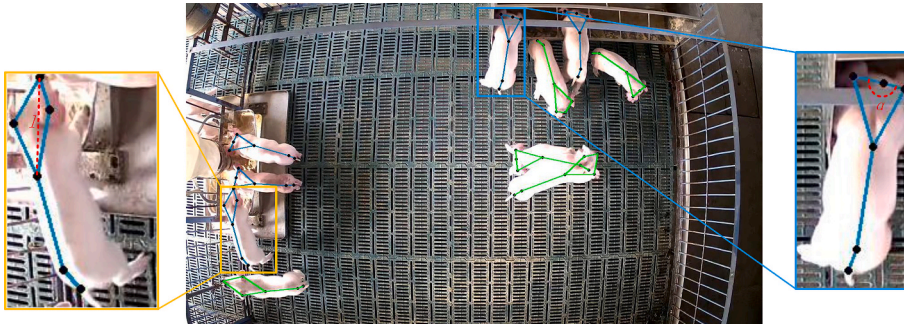


Fig. 8. The keypoint features.

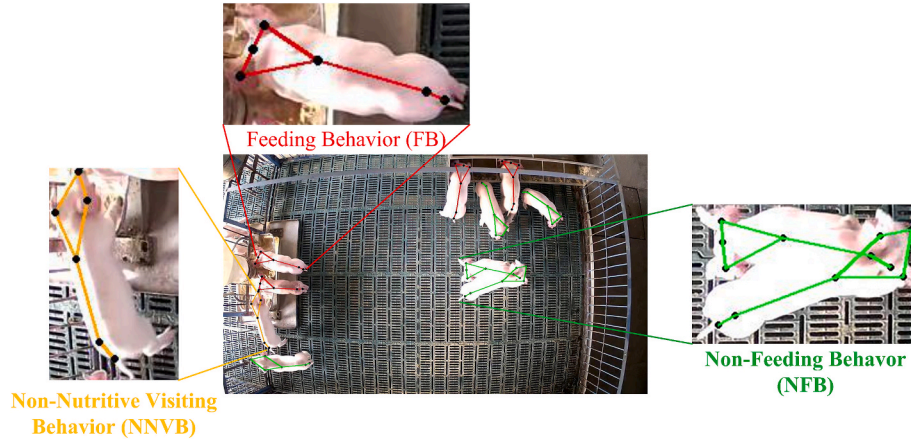


Fig. 9. The three behaviors identified in this article.

Table 4

Experimental environment parameter table.

Name	environment
CPU	13th Gen Intel(R) Core(TM) i7-13700F 2.10 GHz
GPU	NVIDIA GeForce RTX 4090
Memory	24 GB
System	Ubuntu 22.04 LTS
Frame	Pytorch

represents the angle α formed by the Left_ear-Nose-Right_ear, and the red line represents the Euclidean distance l between the Nose and Shoulder. Let H denote the remaining pigs and n denote the dimensionality of the remaining pigs, k initial cluster centers, $C_i (1 \leq i \leq k)$, are selected, and the Euclidean distance $d(H, C_i) = \sqrt{\sum_{j=1}^n (H_j - C_{ij})^2}$ between the remaining pigs and C_i is calculated. Based on the angle and Euclidean distance, the pigs are divided into clusters based on the similarity with C_i . The positions of C_i are continuously updated to minimize the Sum of Squared Errors (SSE) of the clusters. Clustering ends when SSE no longer changes, and the final result is obtained. The formula for calculating SSE is as follows:

$$SSE = \sum_{i=1}^m \sum_{H \in C_i} |d(H, C_i)|^2 \quad (3)$$

Based on the clustering results of K-Means, the optimal hyperplane between different clusters is determined (Hearst et al., 1998). By segmenting different categories through the hyperplane, the clustering result of this experiment is in a two-dimensional space, with the optimal hyperplane being a one-dimensional line. The formula for the optimal hyperplane is as follows:

$$aa + bl + c = 0 \quad (4)$$

In the above formula, α represents the angle of the Left_ear-Nose-Right_ear, and l represents the Euclidean distance between the Nose and Shoulder. a , b , and c represent the parameters of the optimal hyperplane. The obtained optimal hyperplane serves as the criterion for determining the feeding behavior of group-housed pigs, dividing visiting

feeding zone pigs into FB and NNVB.

As shown in Fig. 9, pigs exhibiting FB are marked with red skeletons, pigs exhibiting NNVB are marked with yellow skeletons, and pigs NFB are marked with green skeletons.

2.4. Experiment preparation

The experimental model training environment used in this study is an Ubuntu 22.04 operating system with the hardware configuration shown in Table 4. The experiments were conducted using the Python programming language, and the network architecture, training, and testing were performed using Torch 1.12.0.

In this experiment, all input images of the network are resized to 256×192 pixels. A batch size of 16 images is used, and the weight values are updated and saved once per epoch. The weight decay rate is set to 0.5, and the initial learning rate is set to 0.001.

2.5. Evaluation metric

The evaluation metrics for pose estimation are typically Object Keypoint Similarity (OKS) and Average Precision (AP).

$$OKS = \frac{\sum_i \exp\{-d_i^2 / 2S^2\} \delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \quad (5)$$

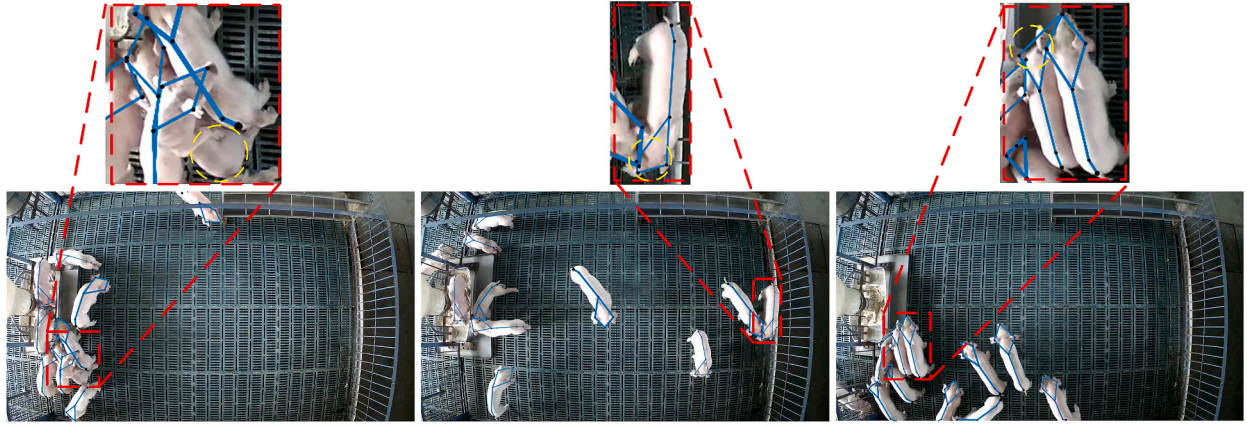
where d_i represents the Euclidean distance between the i -th keypoint's ground truth and predicted values, S represents the square root of the area of the detection bounding box $S = \sqrt{wh}$, w and h are the width and height of the bounding box, σ_i is the standard deviation of the i -th keypoint's manually annotated and actual values, v_i represents the visibility of the keypoint ($i = 1$ indicates that the keypoint is unoccluded and annotated, $i = 2$ indicates that the keypoint is occluded and annotated), and $\delta(*)$ denotes that if the condition $*$ holds true, $\delta(*) = 1$, otherwise, it is 0.

This study uses AP , AP_M , AP_L , AR , AR_M , AR_L as evaluation metrics. AP is the average precision of keypoint accuracy for each detection type when $OKS > T (T \in [0.5 : 0.05 : 0.95])$. AP_M and AP_L represent the accuracy of keypoints for medium-sized and large-sized objects, respec-

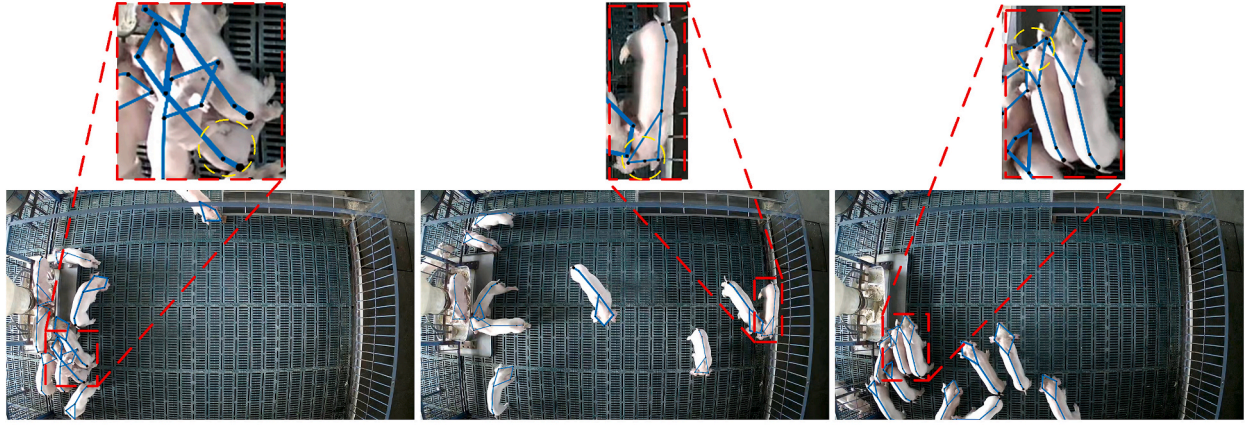
Table 5

Performance comparison of different models for pig pose estimation.

Method	AP	AP_M	AP_L	AR	AR_M	AR_L
KAPAO (McNally et al., 2022)	28.1	0.5	32.9	36.6	13.3	37.2
HigherHRNet (Cheng et al., 2020)	80.8	24.3	83.6	85.6	63.2	86.2
DeepLabCut (Mathis et al., 2018)	84.8	62.3	88.7	87.3	68.4	89.5
HRNet (Wang et al., 2020)	96.6	80.8	97.0	97.5	90.1	97.7
Ours	97.1	83.6	97.5	97.8	91.4	98.0



(a) Pose estimation based on HRNet



(b) Pose estimation based on Pig-HRNet

Fig. 10. Comparison of pose estimation effect between Pig-HRNet and HRNet.

Table 6

The recall rates of pig feeding behavior within the feeding zone for different keypoints.

Keypoints ID	Recall/%	Keypoints ID	Recall/%
[0]	100.0	[0,1,2,3]	84.9
[0,1]	97.3	[0,1,2,3,4]	3.9
[0,2]	2.5	[0,1,3,4,5]	0
[0,1,2]	96.0	[0,2,3,4,5]	0
[0,1,3]	0.6	[0,1,2,3,4,5]	3.4
[0,2,3]	2.0		

tively. In the images, pig regions with an area greater than 32×32 but less than 96×96 are defined as medium-sized objects, while those greater than 96×96 are defined as large-sized objects. AR denotes average recall when $OKS > T$.

We evaluate the group feeding behavior of pigs by calculating Precision (P) and Recall (R). The evaluation formulas are shown in formula (6):

$$P = \frac{TP}{TP + FP}$$

$$R = \frac{TP}{TP + FN} \quad (6)$$

In the equation, TP represents the number of feeding pigs predicted as feeding pigs, FP represents the number of non-feeding pigs predicted as feeding pigs, and FN represents the number of feeding pigs predicted as non-feeding pigs.

3. Results and discussion

3.1. Comparison of pose estimation between Pig-HRNet and different CNN networks

Accurate pose estimation is a prerequisite for recognize group feeding behavior in pigs, so it is necessary to analyze the results of pose estimation in group-housed pigs. To verify the effectiveness of introducing the CotNet Block, this paper conducts a performance comparison of the improved HRNet before and after incorporating the CotNet Block based on a self-constructed pose estimation dataset, and compares them with other commonly used models for pose estimation, as shown in Table 5.

From Table 5, it can be seen that the Pig-HRNet network proposed in this paper achieves the highest AP and AR for medium-sized and large-sized objects. The AP_M , AR_M , AP_L , and AR_L of Pig-HRNet reached 83.6 %, 91.4 %, 97.5 %, and 98.0 %, respectively. Compared to the previous HRNet, these values improved by 2.8, 1.3, 0.5, and 0.3 percentage points, respectively. The increase in AP and AR for medium-sized pigs is more significant because the number of medium-sized pigs in the images is relatively high.

The Pig-HRNet pose estimation model proposed in this paper achieves an AP of 97.1 %, which is 0.5 percentage points higher than HRNet. It also outperforms the KAPAO, HigherHRNet and DeepLabCut pose estimation models by 69.0, 16.3 and 12.3 percentage points, respectively. The AR of the Pig-HRNet model reaches 97.8 %, which is an increase of 0.3 percentage points compared to the improved HRNet. It also outperforms KAPAO, HigherHRNet and DeepLabCut by 61.2, 12.2

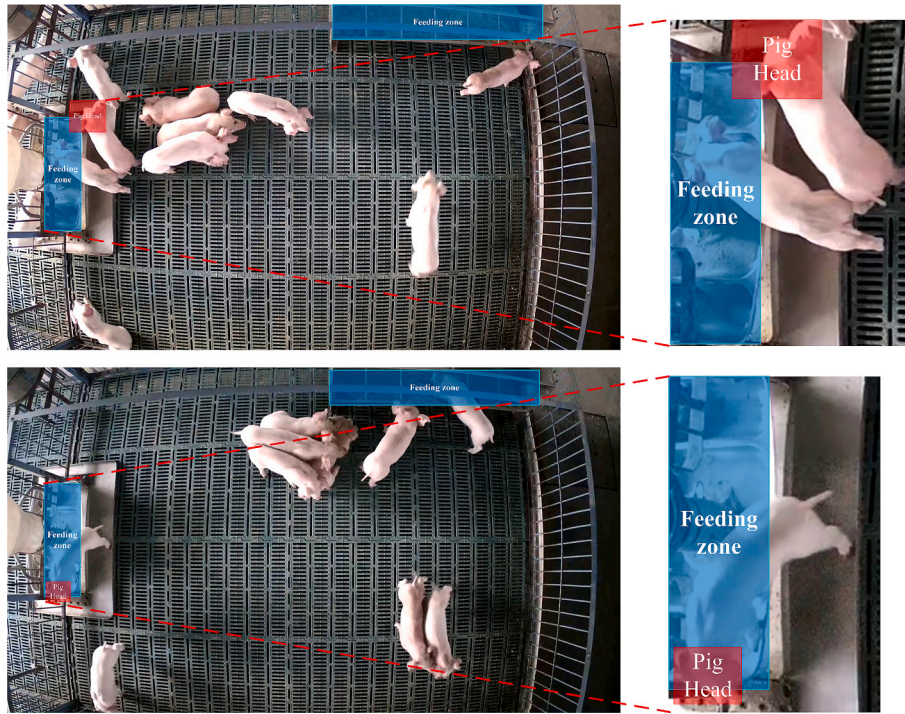


Fig. 11. NNVB is misidentified as FB.

and 10.5 percentage points, respectively. The proposed Pig-HRNet model achieves the highest AP and AR in pose estimation. Compared to models such as KAPAO, HigherHRNet, and DeepLabCut, Pig-HRNet enhances the feature extraction capability of the network by fusing information from multiple resolutions. Additionally, the integrated CotNet Block effectively captures contextual information between adjacent regions and infers the positions of occluded keypoints based on surrounding visible areas. The Pig-HRNet model proposed in this paper performs relatively well in detecting occluded keypoints of group-housed pigs under crowded conditions, overall improving the AP and AR of pose estimation for group-housed pigs.

To more intuitively demonstrate the results of the Pig-HRNet pose estimation model, the pose estimation results of the group-housed pigs obtained from the Pig-HRNet model and the HRNet model are compared, as shown in Fig. 10. Fig. 10(a) shows the visualization of the pose estimation results from the HRNet model, and Fig. 10(b) shows the visualization of the pose estimation results from the Pig-HRNet model. As shown in Fig. 10(a), under crowded conditions, individual keypoints of group-housed pigs may experience missed detections and false detections, especially when the pigs lower its heads, where the occluded Nose keypoint are often misdetected. As shown in Fig. 10(b), the Pig-

HRNet pose estimation model proposed in this paper can accurately detect the occluded keypoints of individual pigs. Accurate keypoint coordinate information is particularly important for subsequent group-housed pig feeding behavior recognition. The misdetected and missed keypoints before and after the improvement of Pig-HRNet are marked with yellow circles in Fig. 10.

3.2. The importance of comparing the relevance of different keypoints with the feeding zone in pigs' behavior of visiting the feeding zone

During the keypoint features discrimination stage, pig behavior is classified into visiting the feeding zone and Non-Feeding Behavior (NFB) based on the relationship between the pig's keypoints and the feeding zone. From the monitoring perspective, when a pig is visiting the feeding zone, the Nose keypoint will always appear within the feeding zone. However, the relationship between the other keypoints and the feeding zone cannot be determined. Therefore, an exhaustive method is used to combine the Nose keypoint with the other five keypoints, calculating the recall rate of pig Feeding Behavior (FB) for each combination.

Table 6 presents the missed detection rates of different keypoints for FB within the feeding zone of pigs. The recall rate of pig FB is 100 %

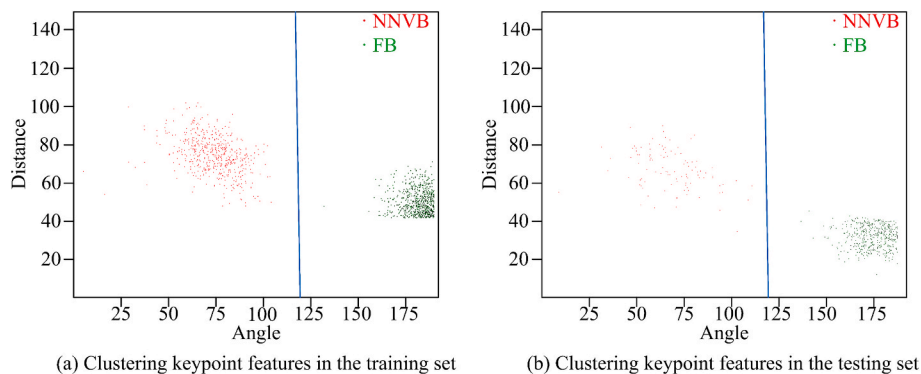


Fig. 12. Clustering visualization.

Table 7

Precision and Recall of three behavior recognition.

Behavior	Precision/%	Recall/%
FB	98.8	99.9
NNVB	99.1	91.3
NFB	100.0	100.0
All	99.3	97.1

when the Nose keypoint is within the feeding zone, indicating that all actual feeding pigs are accounted for in this situation. When the Nose and Left_ear keypoints are within the feeding zone, the recall rate for pig FB is 97.3 %. When the Nose, Left_ear, and Right_ear keypoints are within the feeding zone, the recall rate is 96.0 %. When the Nose, Left_ear, Right_ear, and Shoulder keypoints are within the feeding zone, the recall rate is 84.9 %. For all other combinations, the recall rate for pig FB is below 5 %.

The pigs with the Nose keypoint inside the feeding zone include all feeding pigs. This method also accurately identifies NFB pigs. Therefore, the behavior of group-housed pigs is classified based on whether the Nose keypoint is within the feeding zone. When the Nose keypoint is in the feeding zone, the pigs are categorized as visiting the feeding zone. When the Nose keypoint is not in the feeding zone, the pigs are categorized as NFB. In summary, the relationship between the Nose keypoint and the feeding zone is extremely important for the classification of pig NFB and visiting the feeding zone.

The common approach to recognize pig's FB is to determine whether the head detection box overlaps with the feeding zone. However, in actual production environments, there are cases where the pig's head area intersects with the feeding zone but the mouth is clearly outside the feeding zone, as shown in Fig. 11. The method based on the head detection box fails to focus on the keypoints of the pig, resulting in mis-detection of some pigs. This paper classifies the pigs in the group into visiting the feeding zone pigs and NFB pigs based on whether the Nose keypoint is within the feeding zone. Compared with the detection box, the keypoint is less disturbed by background information and pays more attention to the pig's mouth.

3.3. Classification of group-housed pigs' FB and NNVB based on K-Means clustering

During pig feeding, they typically exhibit a lowered posture, and although the Nose keypoint may be occluded in the image, it can still be detected. Pigs displaying Non-Nutritive Visiting Behavior (NNVB) tend

to lift their heads, with the Nose keypoint unoccluded and clearly visible from a bird's-eye view. Based on these two distinct postures, we use the angle formed by the left ear-nose-right ear and the Euclidean distance between the nose and shoulder as features to cluster pigs accessing the feeding zone. The clustering results, as shown in Fig. 12(a), clearly delineate between FB and NNVB.

The optimal hyperplane calculated after clustering serves as the basis for classifying FB and NNVB. Samples to the left of the optimal hyperplane (red samples) are classified as NNVB, while samples to the right of the optimal hyperplane (green samples) are classified as FB. The formula for the optimal hyperplane is as follows:

$$72.98\alpha + l - 8696.86 = 0 \quad (7)$$

where α is the angle formed by Left_ear-Nose-Right_ear, and l denotes the Euclidean distance from Nose to Shoulder. Using the above formula to classify the FB of pigs in the test set, the classification of keypoint features is shown in Fig. 12(b).

The images of group-housed pigs in this study are divided into three categories: FB, NNVB, and NFB pigs. Table 7 shows the recognition accuracy of the proposed method for these three behaviors. The precision and recall rates for recognizing NFB in group-housed pigs both reached 100 %. The precision and recall rates for recognizing FB were 98.8 % and 99.9 %, respectively, indicating that almost all feeding pigs were correctly recognized. For recognizing NNVB, the precision was 99.1 % and the recall was 91.3 %. The relatively lower recall rate is due to the influence of pose estimation results, where inaccuracies in detecting certain keypoints led to errors in behavior recognition.

3.4. Comparison of pig feeding behavior recognition based on pose estimation and bounding boxes

Currently, researchers recognize the feeding behavior of group-housed pigs based on the method of detecting pig heads using detection boxes. When the pig head detection box intersects with the feeding zone, the pig is considered to be feeding.

Group-housed pig head detection is shown in Fig. 13. Occupation index is used to determine whether a place is covered by a certain object or not. The occupation index of feeding zone are listed respectively in Eq. (8).

$$OcpIndex_{feed} = \frac{AreaofHead \cap AreaofFeedingzone}{AreaofFeedingzone} \quad (8)$$

(Yang et al., 2018) and (Luo et al., 2024) argue that, when

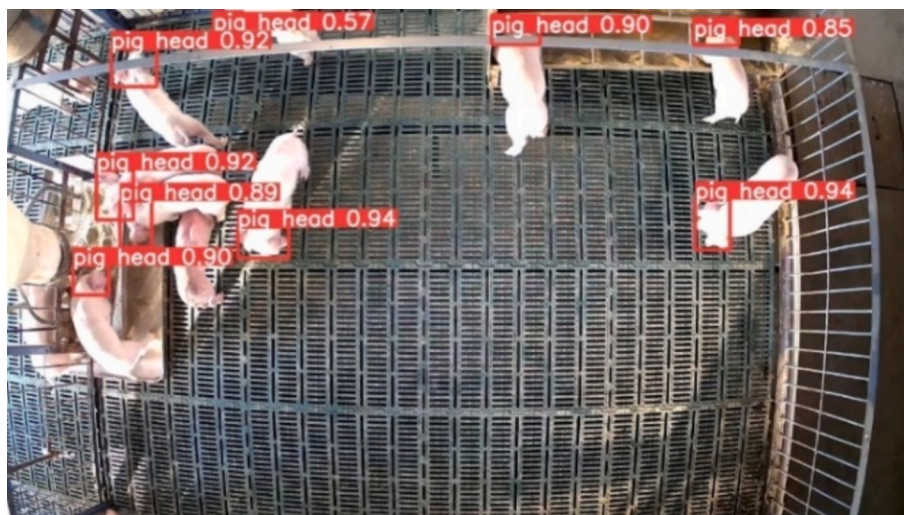


Fig. 13. Group-housed pig head object detection.

Table 8

This paper compares the proposed method with the pig head detection boxes-based feeding behavior recognition method.

Method	Precision/%	Recall/%
pig head detection boxes-based	83.4	99.3
Ours	98.8	99.9

$OcpIndex_{feed}$ is greater than 0, pigs are feeding.

Table 8 presents the results of the pig feeding behavior recognition method based on pose estimation and keypoint features discrimination, as well as the feeding behavior recognition method based on pig head detection boxes, on the dataset used in this study.

The precision and recall rates of the pig head detection boxes-based method are 83.4 % and 99.3 %, respectively. From the surveillance perspective, there are often instances where the pig's head intersects with the feeding zone, but the pig's mouth is outside the feeding zone, indicating Non-Feeding Behavior (NFB). Additionally, there are NNVB where pigs visiting the feeding zone but do not eat. The pig head detection box-based method only focuses on the relationship between the pig's head area and the feeding zone, without considering NNVB and NFB, resulting in a feeding behavior recognition precision of only 83.4 %.

The proposed feeding behavior recognition method in this paper achieves precision and recall rates of 98.8 % and 99.9 %, respectively, both of which are higher than those of the pig head detection box-based method. The method proposed in this paper clearly delineates the feeding behavior of group-housed pigs, eliminating misdetections caused by NNVB and NFB.

The feeding behavior recognition method proposed in this study demonstrates significant advantages. The visualization of feeding behavior is shown in Fig. 14. Fig. 14(a) shows the original image, 14(b) shows the real behavior of the pigs, 14(c) illustrates the feeding behavior recognition based on the pig head detection box, and 14(d) depicts the method proposed in this paper. FB is represented by red rectangular boxes and red skeletons, NNVB by yellow rectangular boxes and yellow skeletons, and NFB by green rectangular boxes and green skeletons. The method based on the pig head detection box does not account for NNVB, resulting in misdetection of NNVB as FB in 14(c). The proposed method classifies pig behavior into FB, NNVB, and NFB, fully considering the impact of different behaviors, thereby accurately recognizing pig feeding behavior. The recognition of the three behaviors in 14(d) is accurate and matches the real behaviors shown in 14(b).

In summary, the pig feeding behavior recognition method proposed in this paper, based on pose estimation and keypoint features discrimination, not only accurately recognizes pig FB but also precisely recognizes NNVB and NFB. This method reduces errors caused by interference

between different similar behaviors during the recognition process, addressing the issue of low precision in pig feeding behavior recognition.”.

3.4.1. Comparison of pig feeding behavior recognition based on Pig-HRNet and existing pig feeding behavior recognition methods

Table 9 illustrates the results of the feeding behavior recognition method proposed in this paper and those reported in existing literature for pigs. In this study, Yang et al. (Yang et al., 2018) utilized the Faster R-CNN algorithm to detect the individuals and heads of group-housed pigs. They achieved feeding behavior recognition by determining whether the head's bounding box intersects with the feeding zone. Tran et al. (Tran et al., 2023) employed the results of YOLOv7 object detection and DeepSORT tracking to determine the behavior of each pig, if the posture of a pig recognized in the detection phase is lying or feeding, its behavior is determined to be the same. If the posture of a pig recognized in the detection phase is standing, the change in its position in the tracking phase (if any) will be used to determine if it is standing or moving. Tu et al. (Tu et al., 2024) used three different detectors from the YOLO series, namely YOLOv5, YOLOX, and YOLOv8, as detectors to compare the effects of behavior classification and tracking, and ultimately chose YOLOv5 with the highest detection accuracy as the pig feeding behavior recognition model.

As shown in Table 9, although the method proposed in this paper did not achieve the highest recognition precision for pig feeding behavior in group farming, it was 0.8 percentage points lower in precision compared to the method proposed by Yang et al. Considering possible reasons, on one hand, the pigs captured in this study exhibited three behaviors: Feeding Behavior (FB), Non-Nutritive Visiting Behavior (NNVB) and Non-Feeding Behavior (NFB). The NNVB was more obvious than the data used in other literature, which increased the difficulty of recognize pig feeding behavior. On the other hand, the average number of pigs in the images used in this experiment was over 10, while the average number in the images used by Yang et al. was around 4, which also increased the difficulty of recognize feeding behavior. Compared to other literature, the method proposed in this paper for recognizing feeding behavior in group-housed pigs achieves the highest recall rate with almost no missed detections of pig feeding behavior. Compared to parameters in other existing literature, the model proposed in this paper for recognizing feeding behavior in group-housed pigs exhibits significant advantages.

In summary, the method proposed in this paper is currently at the forefront of pig feeding behavior recognition. It has achieved excellent results in both precision and recall, meeting the demand for accurate identification of feeding behavior in group-housed pigs under farming conditions.

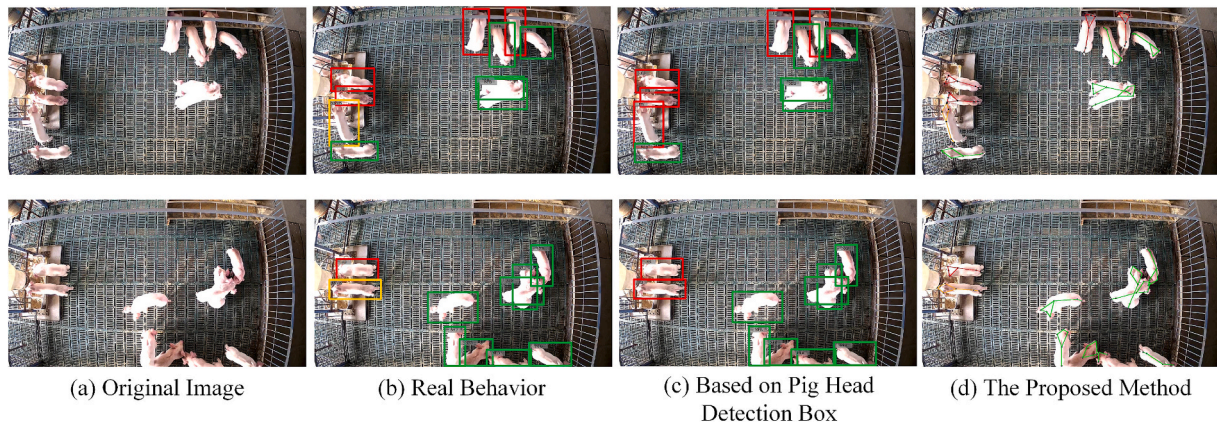


Fig. 14. Visualization of feeding behavior recognition results.

Table 9

The results of the proposed method in this study are compared with the results of existing feeding behavior methods.

Method	number of pigs	data	Algorithms	Precision/%	Recall/%
(Yang et al., 2018)	4	500 images	Faster R-CNN	99.6	86.9
(Tran and Thanh, 2023)	18	3069 images	YOLOv7	97.0	97.0
(Tu et al., 2024)	15	1271 images	YOLOv5-Byte	91.8	89.9
ours	10	1400 images	Pig-HRNet + Kmeans	98.8	99.8

3.5. 24-hour feeding behavior statistics of group-housed pigs

Fig. 15 presents the statistical analysis of feeding behavior of group-housed pigs within 24 h. The x-axis represents the time point, and the y-axis represents the number of pigs that are feeding at that moment. The red dashed line indicates the feeding time points by the caretaker. The method proposed in this study predicts the feeding behavior of group-housed pigs in the pigsty by monitoring and counting the number of pigs feeding every minute. Judging from the number of pigs eating, group-housed pigs tend to eat in clusters, and there are few cases where a pig eats alone, which may be due to their natural gregariousness. The feeding behavior of group-housed varies with time, showing four feeding peaks at 8:15–11:20 GMT + 8, 13:30–15:30 GMT + 8, 15:30–16:30 GMT + 8 and 16:30–17:30 GMT + 8. Before 8:15 GMT + 8, the feeder pours the feed into the feeding zone, and most piglets wake up at this time. During 8:15–11:20, there are continuously multiple pigs feeding, which lasts until before the pigs' noon break. From 11:20 to 13:30, the feeding behavior of pigs declines slightly. During the pigs' midday rest period, the feeder will add feed to the feeding zone. From 13:30 to 15:30, some pigs are sleeping or active, while others wake up and begin feeding. After 15:30, the feeder feeds again, and the feeding behavior of pigs increases. At this time, all the pigs begin to start feeding one after another. After 17:30, group-housed pigs begin to rest and their feeding behavior starts to decrease. Occasionally, there are cases where multiple pigs eat together, but most of the time, most pigs are sleeping, which continues until the next morning. After 18:30, the feeder performs the final feeding, preparing enough food for the pigs that wake up to eat

during the night. Therefore, the feeding behavior of piglets exhibits distinct diurnal rhythmic characteristics and shows intermittent patterns.

4. Conclusion

This paper proposes a method for recognizing Feeding Behavior (FB), Non-Nutritive Visiting Behavior (NNVB), and Non-Feeding Behavior (NFB) in group-housed pigs based on pose estimation and keypoint features discrimination. A pose estimation network, Pig-HRNet, which effectively addresses the issue of misdetection of occluded keypoints in crowded conditions. The pose estimation AP for group-housed pigs reaches 97.1 %. Compared to classic network models such as KAPAO, HigherHRNet, DeepLabCut, and HRNet, the AP of Pig-HRNet improves by 69.0, 16.3, 12.3, and 0.5 percentage points, respectively. By determining whether the Nose keypoint is within the feeding zone, the behaviors of group-housed pigs are classified into visiting the feeding zone and NFB, with both the precision and recall rates for NFB reaching 100 %. Pigs visiting the feeding zone are clustered based on keypoint features. The optimal hyperplane is calculated as the basis for distinguishing between FB and NNVB. The precision and recall rates for FB are 98.8 % and 99.9 %, respectively. The precision and recall rates for NNVB are 99.1 % and 91.3 %, respectively. The feeding behavior recognition method proposed in this paper can accurately identify individual pig behaviors. In this study, four keypoints (Nose, Left_ear, Right_ear, and Shoulder) were primarily used for feeding behavior recognition. However, additional keypoints such as the Buttocks and Tail were also annotated to support future research on more diverse pig behavior categories and to further enhance pose estimation accuracy. In the future, it is necessary to address the impact of pig pose estimation results on behavior recognition and to expand the categories of pig behavior recognition. Additionally, more effective feature extraction modules will be explored to better extract keypoint information features.

CRedit authorship contribution statement

Yuhang Hu: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Xin Dai:** Software, Funding acquisition. **Baisheng Dai:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Data curation. **Ran Li:** Validation, Resources. **Junlong Fang:** Validation, Resources. **Yanling**

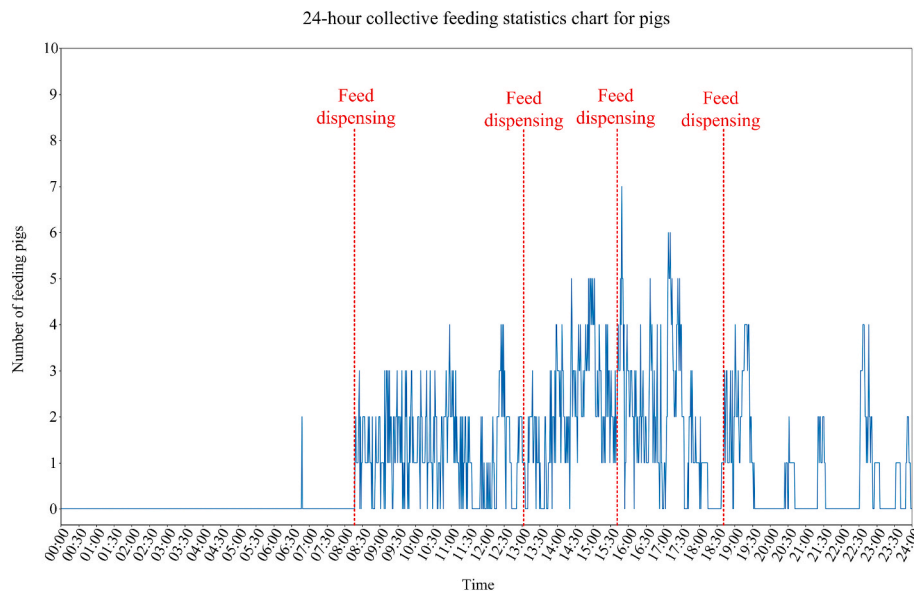


Fig. 15. 24-hour feeding behavior count chart for pigs.

Yin: Validation, Software, Resources. **Honggui Liu:** Funding acquisition, Conceptualization. **Weizheng Shen:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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